

Shosuke Kiami

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EDUCATION **University of California, Berkeley**, Berkeley CA Aug 2025 - Present
Ph.D. Applied Mathematics

University of Washington, Seattle WA Sep 2020 - Jun 2024
B.S. in Computer Science, B.A. in Mathematics
Cumulative GPA: 3.98/4.00
Honors: Magna Cum Laude, Phi Beta Kappa, Annual Dean's List

PUBLICATIONS **Shosuke Kiami**. Lossless Convexification for Linear Systems with Piecewise Linear Controls. [arXiv:2411.18004](#).

- Under review at *American Control Conference 2026*.

Alex Albors, François Clément, **Shosuke Kiami**, Braeden Sodt, Ding Yifan, Tony Zeng. Approximately Jumping Towards the Origin. [arXiv:2412.04284](#).

- Under revision at *Involve Undergraduate Mathematics Journal 2025*.

Chris Jennings-Shaffer, David H Rich, Matthew Macaulay, Michael D Karcher, Tanvi Ganapathy, **Shosuke Kiami**, Anna Kooperberg, Cheng Zhang, Marc A Suchard, Frederick A Matsen IV. Finding high posterior density phylogenies by systematically extending a directed acyclic graph. [arXiv:2411.09074](#).

- Accepted to *Algorithms for Molecular Biology 2024*.

Zoey Chen, **Shosuke Kiami**, Abhishek Gupta, Vikash Kumar. GenAug: Retargeting behaviors to unseen situations via Generative Augmentation. [arXiv:2302.06671](#).

- Accepted to *Robotics: Science and Systems 2023*, awarded Best System Paper Finalist.

Michael Duan, **Shosuke Kiami**, Logan Milandin, Johnson Kuang, Michael Saugstad, Maryam Hosseini, and Jon E. Froehlich. Scaling Crowd+AI Sidewalk Accessibility Assessments: Initial Experiments Examining Label Quality and Cross-city Training on Performance. [dl.acm.org](#).

- Accepted to *ACM SIGACCESS Conference on Computers and Accessibility 2022*.

RESEARCH **Independent Research**, UC Berkeley, Berkeley CA Sep 2025 - Present
EXPERIENCE *Advisor: Jelani Nelson*

- Exploring various applications of stochastic processes to streaming algorithms.
- Developing a new $O(\log n)$ bit super heavy hitter algorithm with success probability $1 - \varepsilon^k$, making progress towards solving ℓ_2 heavy hitter in optimal $O(1/\varepsilon^2 \log(n))$ space.

Independent Research, UC Berkeley, Berkeley CA Sep 2025 - Present
Advisor: Satish Rao

- Independent reading on various aspects of the KLS conjecture, including Bo'az Klartag's new $O(\sqrt{\log n})$ bound on the isoperimetric constant for high dimensional convex bodies.
- Re-derived the proof for the stopping time of Eldan's stochastic localization process.

Independent Research, UC Berkeley, Berkeley CA Jun 2025 - Aug 2025
Advisor: Venkatesan Guruswami

- Developed new exponential-time approximation algorithms for various constraint satisfaction problems such as Max- k Lin.

Autonomous Controls Lab, UW, Seattle WA Jan 2024 - Jun 2025
Advisor: Behçet Açikmeşe

- Single-authored a paper ([arXiv](#)) introducing a new optimal control algorithm that extends lossless convexification theory from stepwise to piecewise-linear parameterized controls.
- Proved guarantees of the above algorithm using results from books such as *Convex Analysis* by Borwein & Lewis and *Numerical Optimization* by Nocedal & Wright.

	Washington Experimental Mathematics Lab , UW, Seattle WA Sep 2024 - Dec 2024 <i>Advisor: Stefan Steinerberger</i> - Co-authored a paper (arXiv) examining a class of dynamical systems in the complex plane. - Discovered and proved results for the stochastic variant, including our main theorem regarding the existence and expectation of the stationary measure in higher dimensions. - Presented at the <i>Northwest Undergrad Math Symposium 2024</i>, receiving 2nd place prize.
	WEIRD Robot Learning Lab , UW, Seattle WA Jan 2023 - Jun 2023 <i>Advisor: Abhishek Gupta</i> Collaborated on a paper (arXiv) introducing generative models as an effective data augmentation technique for robot learning, winning Best System Paper Finalist at <i>RSS 2023</i>.
	Matsen Group , Fred Hutchinson Cancer Center, Seattle WA Jun 2021 - Sep 2021 <i>Advisor: Frederick Matsen</i> - Contributed to a paper (arXiv) introducing the sDAG: a data structure that compactly represents a collection of phylogenetic trees in order to perform Bayesian inference. - Developed fast tree rearrangement algorithms necessary for MCMC steps.
	Makeability Lab , UW, Seattle WA Jun 2020 - Jun 2022 <i>Advisor: Jon Froehlich</i> - Started an initiative to use deep learning for automatic sidewalk accessibility evaluation. - Co-authored a paper (dl.acm.org) examining model performance across international cities.
WORK EXPERIENCE	Software Engineering Intern , Apple, San Diego CA Jun 2023 - Sep 2023 Introduced an improved adaptive bitrate algorithm reducing runtime from $O(n^2)$ to $O(n)$.
	Software Engineering Intern , Microsoft, Redmond WA Jun 2022 - Sep 2022 Designed a custom image segmentation algorithm reducing runtime from 80 ms to 0.5 ms.
TEACHING EXPERIENCE	MATH 54 - Linear Algebra , TA, Berkeley CA Sep 2025 - Dec 2025 - Led 6 hours of weekly discussion sections on linear algebra and differential equations. - Wrote/proctored exams, graded assignments, and held weekly office hours.
	Fusion Math , Math Tutor, Seattle WA Jan 2025 - Jun 2025 Taught lessons covering curriculum in AP Calculus.
	CS Tutor , Self-employed, Seattle WA Mar 2022 - Jun 2022 Taught lessons covering curriculum in AP Computer Science.
	Seattle Music Partners , Cello Teacher Sep 2016 - Jun 2020 Taught cello lessons to students at underserved elementary schools.
PERSONAL PROJECTS	Euclideanate Nov 2024 - Present A highly-optimized C++ algorithm that solves straightedge + compass construction problems, with the goal of finding/proving the minimal construction of the 17-gon (GitHub).
	DanceTime May 2021 - Jul 2023 A C++ <i>Just Dance</i> remake with a custom gesture scoring algorithm using pose estimation, conditional filtering, polynomial regression, and numerical optimization (GitHub).
	AlphaFour Jun 2022 - Jul 2023 A Python <i>Connect 4</i> AI that learns via self-play deep reinforcement learning (GitHub).
	GeoKnowr Nov 2022 - Dec 2022 A Python <i>GeoGuessr</i> AI that reliably guesses within 2000 km of the ground truth (GitHub).
EXTRA-CURRICULARS	UW Formula Motorsports , Engineer Jan 2023 - Jun 2024 Designed the team's first driverless steering system (motor + custom rack and pinion).